

Section A - Multiple Choice Questions (10 × 1 = 10 Marks)

1. Which data structure stores elements in sequential nodes connected by links?

- a) Stack
- b) Queue
- c) Linked List
- d) Tree

2. Which operation adds an element to the top of a stack?

- a) Enqueue
- b) Push
- c) Pop
- d) Insert

3. Which operation removes an element from the front of a queue?

- a) Push
- b) Pop
- c) Dequeue
- d) Delete

4. Which node is present at the top of a tree?

- a) Leaf Node
- b) Parent Node
- c) Root Node
- d) Child Node

5. A graph is made up of _____.

- a) Arrays and Lists
- b) Nodes and Edges
- c) Keys and Values
- d) Parents and Children

6. Which data structure follows the LIFO principle?

- a) Queue
- b) Linked List
- c) Stack
- d) Graph

7. Which data structure follows the FIFO principle?

- a) Queue
- b) Stack
- c) Tree
- d) Linked List

8. In a linked list, each node contains:

- a) Only data
- b) Data and a reference to the next node
- c) Index only
- d) Key only

9. A node with no children in a tree is called:

- a) Root Node
- b) Parent Node

- c) Leaf Node
- d) Branch Node

10. Which traversal explores all neighboring vertices before moving deeper?

- a) DFS
- b) BFS
- c) Inorder
- d) Preorder

Section B - Coding Questions (20 × 2 = 40 Marks)

1. Create a linked list containing the values 10, 20, 30, 40, 50 and display all the elements.

2. Insert a new node containing 25 after the node containing 20.

3. Delete the node containing 30 from the linked list and display the updated list.

4. Count and display the total number of nodes in the linked list.

5. Create an empty stack and push the values 5, 10, 15, 20 into it. Display the stack.

6. Pop one element from the stack and display the updated stack.

7. Display the top element of the stack without removing it.

8. Check whether a stack is empty and display an appropriate message.

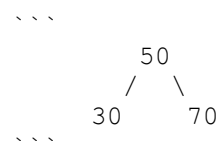
9. Create a queue and insert the values 10, 20, 30, 40.

10. Remove one element from the front of the queue and display the updated queue.

11. Display the front element of the queue without removing it.

12. Display all the elements of the queue in FIFO order.

13. Create the following binary tree:



Display the root, left child, and right child.

14. Perform an Inorder Traversal on a binary tree.

15. Count the total number of nodes present in a binary tree.

16. Find and display all the leaf nodes of a binary tree.

17. Create an undirected graph with the following edges:

- * A - B
- * A - C
- * B - D
- * C - D

Display the adjacency list of the graph.

18. Check whether there is a direct edge between two given vertices.
19. Perform a Breadth-First Search (BFS) traversal starting from vertex A.
20. Perform a Depth-First Search (DFS) traversal starting from vertex A.